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**DEPARTMENT OF COGNITIVE SCIENCE
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**A CLOSER LOOK INTO CROSS-
MODAL CORRESPONDENCES
AND THE ENDOGENOUS
PROCESSES LINKED TO THEM**

Lazarina Georgieva

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Supervisor: Assistant prof. Armina Janyan, Ph.D.

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Lazarina Georgieva


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Abstract

Cross-modal correspondences refer to associations the brain may construct between dimensions of stimuli presented in different modalities. For example, the pitch of a tone (high/low) is associated with the position of a visual object (upper position/lower position, respectively). A number of studies have focused on investigating whether cross-modal correspondences occur in an automatic manner or whether they operate at a more strategic level. Importantly, studies supporting either the automaticity or the presence of an endogenous component in cross-modal correspondences have used very different paradigms to investigate the processes behind the correspondences. The current study aims to specify what those differences are and to fill the disparity in the research by conducting two experiments, focused on investigating how strong is the endogenous component behind cross-modal correspondences. Furthermore, to look closely into cross-modal correspondences by investigating RTs for each of the conditions of tone and position. Results from Experiment 1 replicated previous findings (Evans & Treisman, 2010) and showed cross-modal correspondences between pitch and spatial position, using speeded pitch classification paradigm in auditory modality. Also, results showed that there are asymmetries between the polarities of cross-modal correspondences i.e. cross-modal correspondence effect was significant for tones presented only in the upper spatial condition. Results from Experiment 2, showed that simple probability manipulation (only in the auditory modality) was able to change the cross-modal correspondence effect showing an asymmetrical pattern that seemed to be highly dependent on pitch probability. The results are discussed in light of asymmetrical attentional processes in vertical space.

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Introduction

People experience life through multiple senses. Whether watching TV, listening to music while running through the park or looking for a friend in a crowded room, the brain is constantly managing information from multiple sensory modalities. Since the beginning of the 20th century, psychologists and neuroscientists have conducted studies to investigate how external stimuli from different modalities are processed in the brain, with a rise in the interest on the topic in the last couple of decades.

Some of the research has focused on exploring multisensory integration i.e. the process of selecting, organizing and combining information from different sensory modalities (visual, auditory, touch, smell or taste) into a comprehensive representation is referred to as multisensory integration (Talsma, Senkowski, Soto-Faraco, & Woldorff, 2010). Studies investigating multisensory integration have focused on understanding the factors which might be determinants of which information gets integrated and which processes can modulate multisensory integration (Tang, Wu & Shen, 2016; Busse, Roberts, Crist, Weismann & Woldorff, 2005; Talsma, Senkowski, Soto-Faraco & Woldorff, 2010).

Other studies have focused on cross-modal correspondences. Cross-modal correspondences refer to the associations the brain may construct between the features of stimulus in one modality (e.g. visual) and the features of a stimulus in another modality (e.g. auditory).

For instance, the dimensions of a tone are associated with the size of an object i.e. low-pitched tone is linked to larger objects and high-pitched tone is linked to small sized objects (Parise & Spence, 2012). Thus, in an experimental paradigm, participants are faster to recognize a small sized object when a high-pitched tone is presented simultaneously (Evans & Treisman, 2010). Cross-modal correspondences have been found across senses (touch and vision - Ngo & Spence, 2010b; taste and vision - Spence & Ngo, 2012a), but those are outside of the scope of

current research. The current research will focus on cross-modal correspondences between visual and auditory stimuli.

The following work is structured in a string of accumulating chapters. Chapter 1 will briefly present what multi-sensory integration is, what are the brain areas involved in the process and some important findings concerning the neural mechanisms behind the integration.

Chapter 2 and 3 will discuss in detail audiovisual cross-modal correspondences. Chapter 2 will present in depth what types of cross-modal correspondences have been found, what methodologies have been used and what has been suggested to be the origins of such correspondences. Chapter 3 will focus on processes behind the cross-modal correspondence effect. Specifically, the ongoing debate of whether the cross-modal correspondences are based on exogenous an automatic process, or whether there is an endogenous component involved. Chapter 4 will examine more aspects on the debate on the topic that will lead to the rational of this study. Chapter 5 will present the experiments that have been conducted, following by a general discussion in Chapter 6.

Chapter 1: Multisensory integration

Multisensory integration refers to the process by which information from different sensory modalities is selected and integrated into a comprehensive representation (Tang, Wu & Shen, 2016) and the processes by which information from one modality can impact the processing in another modality (Talsma, Senkowski, Soto-Franco & Wolderff, 2010).

Research on the topic of multisensory integration has shown that it can have an effect on behavior. The influence of multisensory integration on behavior can be separated into two different types (Tang, Wu & Shen, 2016). The first type includes illusory effects, where stimuli from one sensory modality can affect the processing of stimuli in another sensory modality. An example of illusory multisensory effects are the McGurk effect (McGurk & MacDonald, 1976) – where visual information can change audio perception; the ventriloquist effect (Slutsy & Recanzone, 2001), in which the source of a sound is illusory misallocated to the position of a visual stimuli; and the pip-pop effect (Van der Burg, Olivers, Theeuwes & Alais, 2008) in which the location of a visual target is located faster and more accurate, when it is accompanied with a irrelevant “pip” sound.

The other type of effect multisensory integration has on behavior is the performance improvement effect (Tang, Wu & Shen, 2016). That is, results from behavior studies suggest that multisensory integration leads to a more accurate and faster recognition/detection of a stimuli presented in two modalities, compared to when the stimuli is presented in only one sensory modality (Stein, Stanford & Rowland, 2014; Giard & Peronnet, 1999).

But what are the factors that determine which stimuli would be integrated together? Research has found that close presentation in time (Meredith, Nemitz & Stein, 1987), space proximity (Frassinetti, Bolognini & Ladavas ,2002) and intensity of the stimuli (Bolognini, Senna & Maravita, 2010, Meredith & Stein, 1983) to be the major determinants of which two stimuli are going to be selected and integrated together into a comprehensive representation (Talsma,

2015). An argument can be made that these factors only explain the perceptual characteristics of the stimulus. Thus, if these are the main factors that lead to multisensory integration, then it could be argued that multisensory integration is largely stimulus-driven, perceptual processes, rather than high-level cognitive processes. Indeed, in a seminal paper, McDonald, Störmer, Martinez, Feng & Hillyard (2013) discovered that low-level sensory cortical areas considered to be unimodal such as the visual cortex, show activation when participants are presented with only an auditory stimulus (for a review see Störmer, 2019). In four experiments using event-related potential (ERP) recordings, McDonald et al. (2013) found that presenting a task-irrelevant salient sound resulted in activation of the visual cortex. This activation was present even when no visual stimuli were presented to participants, which led the authors to suggest that a salient sound results in spatially specific automatic activation in the visual cortex, even when the sound is irrelevant. It could be argued that the presence of cross-modal activation in the primary sensory cortices, suggests that multisensory integration is an automatic process, mainly dependent on the characteristics of the presented stimuli and not reliant on the context or the task presented.

However, significant research has focused on investigating the role of cognitive factors such as attention in multisensory integration. Based on previous studies which have shown that when attention is directed to a spatial location, processing of all stimuli presented at that location improves, Busse et al. (2005) used ERP and fMRI to further investigate whether these attentional benefits can also spread across sensory modalities, i.e. whether attending to a visual stimulus would benefit the processing of a simultaneously presented auditory stimuli. The results of their study showed that “the brain’s response to task-irrelevant sounds occurring synchronously with a visual stimulus from a different location was larger when the accompanying visual stimulus was attended versus unattended.” (Busse et al., 2005, p. 18751). These results suggest that attention can also contribute to multisensory integration by spread

of attention across different sensory modalities. In another study, Donohue, Roberts, Grent-‘t-Jong & Woldorff (2011) further investigated what is the time window for such spread of activation to occur. In their study, they used the same paradigm of covertly attending to a spatial location where visual stimuli are presented. Their results showed that spread of attention was present when the auditory tone was presented simultaneously (0ms) or with 100ms delay from visual stimuli, but not when the delay was 300ms. So, attention was able to influence multisensory integration, but only when stimulus was presented in a specific time window. Taken together, these results suggest that multisensory integration is dependent on factors such as time, space and intensity, but it can also be moderated by cognitive factors such as attention.

In summary, this chapters aimed to briefly present what multisensory integration is and what are the factors that determine which two stimuli will be integrated together. The next chapter will continue with the topic of cross-modal correspondences. It is important to note, that the difference between multisensory integration and cross-modal correspondences. Multisensory integration describes the brain mechanisms linked to processing stimuli from different modalities, while cross-modal correspondences present associations between the dimensions of such stimuli. Nevertheless, the factors and findings described in this chapter will be further discussed and presented in the context of cross-modal correspondences in following chapters.

Chapter 2: Cross-modal correspondences

2.1 Definition and types of cross-modal correspondences

Cross-modal correspondences have been defined as the associations between the dimensions or features of stimuli presented in different modalities (Spence, 2011). Audiovisual cross-modal correspondences have been discovered between the pitch of a tone and spatial position (Evans & Treisman, 2010; Ben-Artzi & Marks, 1995), size (Evans & Treisman, 2010; Gallace & Spence 2006, Spector & Maurer, 2009), sharpness (Parise & Spence, 2009;), brightness (Martino & Marks, 1999; Mondloch & Maurer, 2004), angle (Parise & Spence, 2012) and spatial frequency of a visual stimulus (Evans & Treisman, 2010). Correspondences have been also found between loudness of an auditory stimulus and vertical motion (Eitan, Schupak & Marks, 2008); loudness and brightness (Marks, 1987a). Additionally, research in cognitive development has found cross-modal correspondences between high pitch tones and small sized bouncing balls in three-year-old children (Mondloch & Maurer, 2004). Also, pre-linguistic 3-4-month infants preferred to look at an elevating ball when it was accompanied with a congruent pitch rising sound of sliding whistle (Dolscheid, Hunnius, Casasanto and Majid, 2014) and Marks, Hammeal & Bornstein (1987) found that 9 years old associate high pitched tones with small sizes and low pitched tones with large sizes.

To put it simply, research has discovered that adults and children tend to associate high-pitched tones with objects positioned higher in space, are smaller in size, with brighter color, sharper edges and acute angles. Low-pitched tones with objects positioned lower in space, are bigger in size, with darker colors, round edges and blunter angles. The next section and chapters will further discuss how have cross-modal correspondences been investigated though the years, what processes are linked with them and suggestions as to how they are formed.

2.2 Investigation of cross-modal correspondences

The majority of studies investigating audiovisual cross-modal correspondences measure reaction times (RT) using the speeded classification paradigm, known as Garner's speeded classification (Garner, 1974; for review Algon & Fitousi, 2016).

In the classic speeded classification tasks, participants are presented with two stimuli, with each stimulus having two or more dimensions (Algon & Fitousi, 2016). Participants are asked to judge the one of the dimensions of a stimulus (for instance, participants could discriminate the color of a square as red or blue) while ignoring the other dimensions of the stimulus (light blue or dark blue), making the color task-relevant and brightness task-irrelevant. Garner's speeded classification task included 3 parts. In the first part, called the baseline, both of the task-relevant dimensions are presented (red or blue), while from the task-irrelevant dimensions only one is shown (dark) (Algon & Fitousi, 2016). For example, when judging the color of the target, the brightness dimensions are a constant (all trials consist of light blue or light red stimuli). In the second part, called the filtering, participants are randomly presented with the dimensions of all stimuli i.e. participants judge color (red vs. blue) while ignoring brightness (dark vs. light). In the third part, called correlation, participants are again presented with the dimensions of all stimuli, but this time the dimensions are correlated on most of the trials i.e. visual stimuli is almost always presented as dark blue vs. light red in one block of trials and light blue vs. dark red in another block of trials.

According to Garner (Garner, 1974), if the two stimuli share integral dimensions, mean RT in the baseline tasks will be faster compared to mean RT in the filtering task (Garner, 1974). This difference in reaction times is known as Garner interference (Algon & Fitousi, 2016). Garner interference can be explained in terms of selective attention, particularly - it is harder for participants to selective attend to the task-relevant dimensions while the task-irrelevant dimensions vary, compared to when they are held constant, resulting in poorer performance

(Algon & Fitousi, 2016). In contrast, if mean RT in the baseline and in the filtering are similar or equal, the two stimuli share separable dimensions. If stimuli have separable dimensions, participants find it easy to ignore the variance in the dimensions (Algon & Fitousi, 2016). Lastly, if the two stimuli share integral dimensions, RTs in the correlation task should be faster than RTs in the baseline. This difference in RT is known as a redundancy gain (Algon & Fitousi, 2016).

Here it is important to mention that most of the studies using Garner's speeded classification paradigm, to investigate whether the dimensions of an auditory and visual stimuli are separable or integral have a few differences. Most importantly, the task relevant dimensions are presented in one modality and the task-irrelevant dimensions are presented in another modality. Furthermore, in the baseline task stimuli are presented in only one modality. For instance, if the task is to classify the position of a visual stimuli presented on screen, no stimuli from other modality (auditory) is presented. Thus, when in the filtering task participants are randomly presented with both task-relevant and task-irrelevant stimuli from different modalities, slower mean RTs, suggest that the dimensions of the two stimuli share integral dimensions, which implies that cross-modal correspondences are present between the features of the stimuli. Thus, participant have trouble ignoring the task-irrelevant dimensions of the stimuli, due to cross-modal correspondences between the features of the stimuli presented in two modalities and they perform poorly in the filtering task, compared to the unimodal baseline task. Cross-modal correspondences through Garner interference has been found between lightness and pitch (Martino & Marks, 1999); visual position and pitch (Bez-Artzi & Marks, 1995) and between loudness and visual vertical motion (Eitan, Schupak & Marks, 2008).

Another type of method used to investigate cross-modal correspondences is though congruency effect. In the context of audiovisual cross-modal correspondences - a congruent trial consists

of a trial introducing stimuli that are suggested to have cross-modally correspondent dimension. For example, high pitch tone and a visual stimulus presented above central fixation, is smaller in size, is brighter and with sharper edges; and incongruent trial will consist of a high-pitched tone and a visual stimulus presented below fixation, is bigger in size, is dimmer and has a rounder edge. Thus, in paradigms investigating the congruency effect - mean RT of congruent trials is compared to mean RT of incongruent trials. If cross-modal correspondences exist between stimuli from different modalities, mean RT for congruent trials should be faster than mean RT for incongruent trials. This would suggest, that performance was better on congruent trials, due to these formed associations between the features of stimuli in auditory and visual modality. Furthermore, congruent and incongruent trials are compared to a control condition in which only one the task relevant stimulus is presented unimodally.

Congruency advantage (mean RT congruent trials < mean RT incongruent trials) of audio-visual cross-modal correspondences have been explored through diverse methodologies. For example, Evans & Treisman (2010) used speeded classification task and found congruency advantage between pitch and visual position (high pitch/upper spatial position; low pitch/lower spatial position), pitch and size (high pitch/smaller objects; low pitch/larger objects), and pitch and spatial frequency (high pitch/high spatial frequency; low pitch/lower spatial frequency). In another study, Gallace & Spence (2006) found congruency advantage between size and tone (high pitch/smaller objects; low pitch/larger objects) using a speeded discrimination paradigm, in which participants compared the size of two disks presented one after another. Congruency advantage between the size and pitch have also been found using speeded implicit associations tasks (Parise & Spence, 2012) and using a temporal order judgment (TOJ) task (Parise & Spence 2009). The existence of Garner interference and congruency effect in all these various paradigms suggest that the associations between dimensions of auditory and visual stimuli exist

and that they are robust enough to be visible in a variety of tasks. The next section will aim to briefly present some suggestions as to how do these correspondences have been formed.

2.3 Origins of cross-modal correspondences

The vast and diverse research on the topic of cross-modal correspondences has shown the various associations the brain makes between dimensions of auditory and visual stimuli. But how are these cross-modal correspondences formed? According to Spence (2011), there are three different classes of cross-modal correspondences – structural, statistical and semantic cross-modal correspondences.

Structural cross-modal correspondences refer to those correspondences that could be the result of neural mechanisms and more specifically – the linked dimensions of sensory stimuli might be processed in nearby regions, or they might be processed using related mechanisms (Spence, 2011). The cross-modal correspondence between loudness and brightness might be categorized as a structural correspondence (Spence, 2011).

Statistical cross-modal correspondences refer to those correspondences that have been formed based on consistencies existing in the surrounding world and previous knowledge. This class of cross-modal correspondences might be explained in terms of Bayesian models and coupling prior (Parise & Spence, 2009). Therefore, features of auditory and visual stimuli that have been regularly seen and experienced together will lead to strengthened connection between the two. Spence (2011) defined pitch-elevation, loudness-size and pitch-size correspondences as statistical and defines them as “learned” (Spence, 2011, p. 987). So, for example, high pitch is associated with small size, because in the surroundings smaller objects/people often have a more high-pitched voice, while larger objects/people tend to have deeper (low) voice.

Considering results from Dolscheid, Hunnius, Cassasanto and Majid (2014) paper, in which pre-linguistic 3-4-month infants preferred to look at an elevating ball when it was accompanied with a congruent pitch rising sound, it could be argued that such statistically learned associations can be formed rapidly during the first months of cognitive development.

Semantic correspondences refer to those correspondences which origin through language development. These correspondences are the result of common labels used in language. For example, pitch-elevation and pitch-spatial frequency have been defined as semantic correspondences since, in the English language for example, “high” and “low” are used to describe both spatial position and pitch of a tones.

Although some research has supported the notion of some of the three classes of cross-modal correspondences (Spence & Deroy, 2013; Parise & Spence, 2009; Ramachandran & Hubbard, 2001) it is important to note that such classification is far from being accepted as the norm. Firstly, Spence (2011) suggest that such classification does not provide explanation for all of the cross-modal correspondences found (for example associations between pitch and brightness currently remain unexplained). And secondly, the pitch-elevation correspondences have been defined as both statistical and semantic (Spence, 2011). Nevertheless, the classification presented in this paper aims to provide a simple account of the origins of cross-modal correspondences presented in the literature.

In summary, this chapter provides general and methodological information concerning cross-modal correspondences. Definition, along with examples of the different type of cross-modal correspondences are provided. In addition, the chapter aims to sum up the different paradigms (speeded classification tasks, implicit association tasks, temporal order judgment tasks,

speeded discrimination tasks) used to investigate associations between features of auditory and visual stimuli, thus showing that cross-modal correspondence exist and can affect performance in a variety of tasks. The next chapter will further explain the processes behind the cross-modal correspondences, specifically the ongoing debate of whether cross-modal correspondences occur in an automatic manner, or whether they are susceptible to endogenous processes.

Chapter 3: Processes linked to cross-modal correspondences

Audiovisual cross-modal correspondences have been found between a number of auditory and visual dimensions. Nevertheless, there is little consensus of the processes behind these associations. One of the most debated questions in recent years on the topic of cross-modal correspondences is whether these associations occur in automatic(exogenous) manner, or whether top-down, endogenous processes also contribute to them. A number of studies have been conducted, using different paradigms and manipulations. The following sections will present in detail studies supporting the automaticity of cross-modal correspondences and studies supportive of the endogenous account behind the effect.

3.1. Studies supporting the automaticity of audiovisual cross-modal correspondences

Initially, cross-modal correspondences were linked to exogenous, automatic processes i.e. it was supposed that congruent trials were classified quicker than incongruent, because in incongruent trials “the irrelevant feature affects the processing of the relevant feature in a bottom-up fashion” (Getz & Kubovy, 2018, p.101).

Evans and Treisman (2010) conducted 9 experiments on cross-modal correspondences between pitch and spatial location, size, spatial frequency and contrast. Participants were presented with direct and indirect tasks. In the direct task participants heard a high/low pitch (1500Hz and 1000Hz respectively) tone for 120ms. Simultaneously, a visual stimulus (circles with black and white sinusoidal grating) was presented above or below fixation. In the direct task, participants needed to discriminate the tone or the spatial position, size, contrast of spatial frequency of the circles. The results showed a cross-modal correspondence effect resulting in participants answering significantly faster to congruent pairs (high pitch tone/ upper spatial

position) than incongruent trials (high pitch/lower spatial position). In the indirect task, participants were asked to discriminate whether the presented tone was piano/violin sound, or whether the grating of the visual stimuli is oriented left/right, while still appearing above/below fixation. Results in the indirect task showed a significant cross-modal correspondence effect, too, in position, size and spatial frequency leading the authors to suggest that cross-modal correspondence occur in automatic manner and do not depend on the task (Evans and Treisman, 2010).

The study conducted by Evans & Treisman (2010) provides some of the strongest support of the notion that cross-modal associations occur in an automatic manner, mainly based on the results shown in the indirect tasks. The paradigm and manipulations used by Evans and Treisman (2010) are simple and refined and show that congruency effects can emerge independent on the task. Nevertheless, it can be argued, that although the paradigm used suggest automatic processes, it fails to show that endogenous processes cannot modulate the congruency effects.

In another study, Parise & Spence (2012) investigated cross-modal correspondences between pitch-size and pitch-size of visual angle (Experiment 3 and Experiment 4 respectively). The visual stimuli in Experiment 3 included 2 grey disks (2cm vs. 5 cm) presented centrally on screen. The auditory stimuli included two pure tones (300Hz. vs. 4500 Hz.). Experiment 4 used the same auditory stimuli (pure tone – 300 vs. 4500Hz.), while the visual stimuli included two angles – one more acute than the other (42° vs. 126°). Participants were presented with either the visual or the auditory stimulus, both of which were presented for 300ms. Participants needed to make an implicit association regarding the size, the angle of the visual stimulus or the pitch of the tone by pressing two keys on a computer keyboard. Participants received feedback for every incorrect answer. Each participant went through 4 blocks of trials repeated

6 times. A congruent block of trial was considered one in which participants needed to respond with the same button for small sized disks and high-pitched tones, or acute angle and high pitch. An incongruent block of trial was considered to be one in which participants needed to respond with the same button for small sized disks and low-pitched tones, or acute angle and low pitch tone. Results from Parise & Spence's (2012) study showed a significant cross-modal congruency effect for both experiments i.e. people were significantly faster to classify small sized disks and acute angles with the same button as they classify high pitched tone. The authors conducted an additional bin analysis. Responses were separated into 5 bins depending on the reaction times – from fastest to slowest. Results showed that participants were faster to the congruent than incongruent trials in every bin. The faster responses to congruent trails compared to incongruent trials even in the fastest reaction times bin, led the authors to suggest that cross-modal correspondences occur automatically. Moreover, given that in an implicit association task, each of the stimulus is presented unimodally, the authors implied that cross-modal correspondences not only happen in an automatic manner, but that multisensory integration can be excluded as an essential mechanism behind the correspondences (Parise & Spence, 2012). Nevertheless, it can be argued that results are not sufficient to support the notion that cross-modal correspondences are an automatic process. For instance, due to the nature of the implicit association, the advantage found for congruent trials, do not necessarily mean that congruency effects are due to cross-modal correspondences. Rather, they can be the product of any other internal associations between the response mapping and the presented stimuli (Parise & Spence, 2012).

Lastly, Mossbridge, Grabowecky & Suzuki (2011) investigated whether cross-modal correspondence between sound (ascending vs. descending) and spatial position can affect performance by involuntarily directing attention to a spatial location using a go/no-go task.

Participants in Mossbridge, Grabowecky & Suzuki (2011) study needed to respond whether the color of a centrally presented circle (referent) matched the color of sequentially presented one (probe). In one of the experiments (Experiment 1) the probe was positioned on a cardinal axis i.e. in squares positioned above, below, left and right to fixation. In another experiment (Experiment 2 and 3) the sequential circle was presented on a diagonal axis i.e. in squares positioned in top left, right and bottom left, bottom left and right of fixation. The referent square was presented for 500ms, along with four counterbalanced tones – 2 ascending (300 to 450 Hz, 450 to 600 Hz.) and 2 descending (450 to 300Hz, 600 to 450 Hz) tones. Afterwards the probe circle was presented at cardinal or diagonal position for 750ms. Participants were instructed to ignore the tones and the position of the probe disk and to respond only to the color i.e. a go task was one in which the color of the referent and the second circles matched (75% trials), a no-go task was one in which the colors were a mismatch. Results from Experiment 1 showed that participants were significantly faster to respond to the circles matching the color of the referent positioned above fixation, when the referent was accompanied with an ascending tone then descending tones. Interestingly, the horizontal axis showed that responses to the right were faster after ascending tones then descending and vice versa for the left location. Usually, results from studies supporting congruency advantage between pitch and horizontal axis, attribute the effect to participants' musical knowledge and background (Deutsch, Hamaoui & Henthorn, 2007).

Furthermore, results from Experiment 2, where the second circle was presented at diagonal location showed no significant effect. That is why, the authors conducted Experiment 3, where the trials were separated into blocks, consisting two of possible locations i.e. 1 block consisted of top and down positions (vertical condition), 2nd block consisted of top left and bottom right positions (diagonal condition), 3rd block consisted of top right and bottom left positions (diagonal condition). The results from the experiment showed a significant congruency effect

only for the vertical condition i.e. RTs were faster when the referent circle was presented with an ascending tone, followed by a target circle at top position. The authors interpreted the results from Experiment 2 and 3 to mean that cross-modal associations between ascending/descending sound and spatial position are associated only with vertical position, which is consistent with other studies (Eitan & Granot, 2006, Deutsch, Hamaoui & Henthorn, 2007). Results from Experiment 1, where congruency effects influenced participants performance, led the authors to suggest that “cross-modal attention effects are likely automatic rather than volitional because, the sounds were uninformative about the upcoming probe location, spatial information was irrelevant to the color-matching task, and the participants were instructed to ignore the sounds” (p. 138). However, it could be argued that such assumption might be rather weak. Firstly, the paradigm used in Mossbridge, Grabowecky & Suzuki (2011) experiment, focuses on investigating direction of spatial attention due to cross-modal correspondences, rather than the processes behind the associations. Secondly, the fact that effect was found between ascending/descending pitch and horizontal axis (an effect usually seen in studies with musicians as subjects), could mean that both the stimuli and the participants background could have been beneficial to the results in the study.

In summary, this section aims to introduce diverse studies, the results of which imply that cross-modal correspondences occur in an automatic manner. The main argument for automaticity derived from the presented studies is that cross-modal correspondences affect participants’ performance in tasks where one of the sensory stimuli is task-irrelevant and even in tasks where the cross-modal dimensions of the two stimuli are not important or informative of the task itself (the indirect tasks Evans & Treisman, 2010; color-matching Mossbridge, Grabowecky & Suzuki (2011)).

Nevertheless, another line of research has implied that cross-modal correspondences do not occur in explicitly automatic manner. The next section will present some of these studies.

3.2 Studies suggesting that endogenous processes can influence audiovisual cross-modal correspondences

In 2012, Chiou & Rich (2012a) conducted 4 experiments to further explore cross-modal correspondences effect on attention allocation. Chiou & Rich (2012a) used a non-informative auditory cue followed by various SOA's and a speeded detection of visual stimuli positioned above/below fixation.

In the 1st experiment, participants heard unpredictable sinusoidal high (1500Hz) or low (300Hz) tone with a presentation time of 200ms, followed by a randomly presented delay time (0ms, 80ms, 160ms, 320ms, 640ms); and finally, a low contrast visual target (circle) presented above/below fixation. On 8% of the trials no visual stimulus was presented (catch trials). Participants were instructed to ignore the sound and to respond only when they see the target – a grey circle. Experiment 1, used vertical and horizontal positions i.e. the circle was presented above or below fixation in one block or left or right to fixation in another block of trials. Results showed a significant effect of Axis (vertical vs. horizontal) and significant interaction between delay time and congruency. The cross-modal congruency effect was present only for the vertical axis, suggesting that pitch and position correspondences are present only at a vertical axis, similar to other studies (Eitan & Granot, 2006; Deutsch, Hamaoui & Henthorn, 2007). The effect size in milliseconds was larger at the longer delay time (640ms), suggesting endogenous processes involved (Chiou & Rich, 2012a).

Of interest Chiou & Rich (2012) conducted another experiment, which is of interest to the current study – Experiment 4 which consisted of two parts of Chiou & Rich (2012a). In

experiment 4A the authors changed the presentation of the tone to 50ms, following duration conventionally used in classic attention orienting paradigms (Chiou & Rich, 2012a). Additionally, in experiment 4B, they manipulated the natural mapping of the cross-modal correspondences and changed the probabilities of the presented trials. In 80% of the trials high/low tone was linked to below/above spatial location i.e. the opposite direction of the natural mapping. Importantly, participants were informed of the probabilities. The delay time presented were 0ms, 150ms, 300ms, 450ms and 600ms for experiment 4A and 0ms, 300ms, 600ms and 900ms for experiment 4B. Results from experiment 4A showed main effect of delay time and congruency. Results of 4B also showed a significant main effect of delay time and congruency i.e. participants responded faster to incongruent trials (high pitch, below fixation spatial location) i.e. participants were able to change the mapping based on instruction. Since there was no significant difference at 0ms delay time, the authors implied that cross-modal correspondences are susceptible to endogenous control, but it takes time (at least 200ms) for the effect to develop and that correspondences can occur at a “relatively late stage of voluntary attention orienting” (Chiou & Rich, 2012a, p.339).

Nevertheless, two main concerns can be raised from the experiments conducted by Chiou & Rich (2012a). Firstly, it could be argued that the experimental procedure used by the authors doesn't measure the processes behind cross-modal correspondences. Instead, the results showed that based on the auditory cues, attention was oriented cross-modally to a congruent location to the pitch of the tone. So, the endogenous processes involved, might be the product of cross-modal correspondences, but they do not account for how cross-modal correspondences are formed. Secondly, there was no significant difference at delay time 0ms, despite the strong probability manipulations used. Even though, probability manipulations were presented across the two sensory modalities (high pitched was linked to a spatial position on 80% of trials), and participants were explicitly informed about such manipulations, no significant difference was

present at 0ms delay. These results, can be explained by the fact that in all experiments the visual modality was task-relevant. Research has suggested, that the visual modality is dominant (Colavita, 1974), so it could be that when presented with stimuli simultaneously, participants were able to focus on the visual task and ignore the auditory cue. Therefore, a question can be raised that is of interest to the current research – will the same results be present in a paradigm where all of the stimuli are presented simultaneously and the task-relevant modality is the auditory one. Furthermore, will the endogenous processes be visible in a paradigm that uses less explicit manipulations. Those questions are one of the central goals of the conducted research and will further be discuss in Chapter 5.

In another study, Klapetek, Ngo & Spence (2012) investigated whether an uninformative auditory tone will affect participants performance in a visual search task, specifically the visual search task used by Van der Burg, Olivers, Theeuwes & Alais (2008) known as the pip-pop effect. Participants needed to respond to a horizontal or a vertical target among tilted distractors (23, 35 or 47 distractors). In order to induce a cross-modal correspondences, the distractors changed brightness (light grey to dark grey) approximately every 900ms. Simultaneous to the appearance of a target, a sine-wave tone (low pitch 250 Hz. tone, high pitch 2000 Hz. tone) was presented on 2/3 of the trials. The rest 1/3 didn't present a sound. Thus, a congruent trial consisted of a low-pitched tone presented before a dark grey target appearing on screen. Participant task was to detect the target and to specify whether its orientation was horizontal or vertical, by pressing two keys on a computer, while ignoring the tone or the brightness change of the stimuli.

Klapetek, Ngo & Spence (2012) conducted a second experiment using the same stimuli and procedure, but this time participants were informed about the cross-modal correspondences between pitch of the tone and brightness of the visual stimuli. Additionally, trials were not,

presented in a randomized order but were separated in blocks – one block consisted of congruent trials, one of incongruent trials and in one no sound was presented, similarly to speeded classification paradigm defined by Garner (Garner, 1974). This could have resulted beneficial to the learning of such associations and consequently the results. At the beginning of each trial participants were informed about the pitch-brightness mapping. Results showed that participants detected and discriminated the targets faster after they were presented with a congruent sound, then incongruent or with no sound. Thus, the authors suggested that cross-modal correspondences between pitch and brightness occur at a more strategic level than previously thought.

The results from the study conducted by Klapetek, Ngo & Spence (2012) can be counted as intriguing because of their difference, due only to explicit instructions provided to participants. So, if we look at the results from Experiment 1, results showed that participants responded significantly faster to target, when an auditory tone was presented simultaneously, but no effect of congruency was shown. The results did show that participants were faster to detect the targets when sound was presented (congruent or incongruent), which is in line with multisensory integration i.e. the mere presence of a sound facilitated target detection. Nevertheless, no congruency effect was found between pitch and brightness. Perhaps, the cross-modal mapping between pitch and brightness is not strong enough to manifest itself in the context of the paradigm used by Klapetek, Ngo & Spence (2012). On the other hand, results in Experiment 2, showed not only better performance when a tone was presented, but it showed congruency effects between pitch and brightness. Since, the only difference between the two experiments was the provided instructions, it can be suggested that the instructions alone resulted in participants increased attention to cross-modal mappings, resulting in stronger associations and faster detection times. The fact that instructions alone, were needed to give rise to such effect, only proves how important they are and how they can affect participants

performance on a task. So, can endogenous processes be introduced in a less implicit manner i.e. without presenting explicit instructions to participants? The current research aims to provide answer to such question.

In summary, this section aims to present studies supporting the notion that cross-modal correspondences can be defined by endogenous processes. Nonetheless, a new view emerged in last couple of years – a recent research suggested that cross-modal correspondences are defined by both exogenous and endogenous processes (Getz & Kubovy, 2018). The next section will briefly review this rather new view.

3.3 Cross-modal correspondences are defined by both exogenous and endogenous processes

In 2018, Getz & Kubovy conducted a study that aimed to investigate the “top-down-ness” and “bottom-up-ness” of pitch and size, elevation, spatial frequency, angularity and brightness. To do so they used a speeded classification paradigm, where congruency and compatibility via instructions are manipulated. Congruency was defined by associations found in previous research between auditory and visual stimuli. Compatibility was defined by the instructions presented to participants. In a compatible block of trials, the instructions given to participants matched the natural congruency of features between auditory and visual stimuli. In an incompatible block of trials, instructions were reversed to the natural mapping e.g. participants were informed that high pitch was linked to large shapes, low pitch is linked to small shapes. The authors hypothesized that if cross-modal correspondences occur in automatic manner results on incompatible trials will be slower than those of compatible trials, suggesting that the reverse mapping of the associations will be difficult to pair. On the other hand, if participants managed to quickly reverse the natural mapping based on the instructions provided, this would

suggest a top-down influence on cross-modal correspondences. Another case the authors hypothesized was to suggest that both bottom-up automatic and top-down processes are involved in cross-modal correspondences. In order to support this case, results would have need to show that participants were faster on congruent trials in the compatible condition and incongruent trials in the incompatible condition. (Getz & Kubovy, 2018). Results suggested that “top-down influence was present to varying degrees in all five cases, with the strongest influence of size and the weakest on height” (Getz & Kubovy, 2018, p. 107). The results led the authors to suggest that both automatic and endogenous processes are involved in cross-modal correspondences.

Chapter 4: Terminological and methodological issues, and rationale for current study

While research on the topic of cross-modal correspondences has been extensive, especially in the last couple of decades, there are still a number of issues that can be raised (for a review see Parise, 2015; Deroy & Spence, 2013). The next sections will briefly present some terminological and methodological concerns regarding the studies supporting exogenous or endogenous processes behind cross-modal correspondences, and how these issues pointed to the rationale behind the current study.

4.1 What defines automaticity

The contradictory results from studies, concerning the automaticity of cross-modal correspondences (Evans & Treisman, 2010; Klapetek, Ngo & Spence 2012; Chiou & Rich, 2015, Parise & Spence, 2012) raise the question of what actually defines automaticity. If we consider results from Evans & Treisman (2010) study, automaticity is defined by the fact that participants responded faster to congruent stimuli even when the dimensions of these stimuli were task independent (Evans & Treisman, 2010). Parise & Spence (2012) suggested that cross-modal correspondences occur in an automatic manner, because cross-modal associations influenced participants performance even in the fastest responses.

On the other hand, Getz & Kubovy (2018) point out three reasons why a congruency advantage is not enough to consider a process being automatic: 1) There are studies that have failed to show congruency advantage for audio-visual cross-modal correspondences (Heron, Roach, Hanson, McGraw & Whitaker, 2012; Sweeny, Guzman-Martinez, Ortega, Grabowecky & Suzuki, 2012); 2) the “congruency advantage fails to show that the effect is immune to endogenous processes” (Getz & Kubovy, 2018, p.101) and 3) there is a very little agreement in what exactly defines automaticity (for a review see Spence & Deroy, 2013). Moors & De

Houwer (2006) defined automaticity as an “umbrella” term which consists of 4 criterions – goal-independence criterion, non-conscious criterion, the load-intensity criterion and speed criterion (Moors & De Houwer, 2006). In order for a process to be automatic, according to the goal-independent criterion, the participants cannot stop or change it from happening i.e. it is not dependent on participants cognitive control. The non-conscious criterion describes a process as automatic when it occurs without the participants conscious awareness i.e. subjects are not consciously aware of the association or the processes occurring. The load-intensity criterion states that in order for a process to be automatic, it needs to not be affected by increase in perceptual load. The speed criterion defines a process automatic, if it occurs at earlier stages of information processing (Spence & Deroy, 2013).

Thus, answering the question of whether cross-modal correspondence effect is an automatic process becomes rather unclear and difficult, especially if one considers the different criteria of automaticity (for instance, should a process be called automatic if it responds to only 1 or 2 criteria), along with the number of cross-modal correspondences found between features of sensory modalities (for example, some mappings might be the product of stronger automatic processes, while other might be more prone to cognitive control), the diverse paradigms used to investigate these associations and the difference level of stimuli salience used in these paradigms.

4.2 Disparity in the research and rationale for current study

The methodology used in the current research aims to fill a disparity in the studies used to investigate processes behind cross-modal correspondences, particularly the level of automaticity of the correspondences. One of the main gaps in the research is that the studies described in previous sections (Chiou & Rich, 2012a, Mossbridge, Grabowecky & Suzuki, 2011, Chiou & Rich 2015), which have suggested that cross-modal correspondences do not

occur in a fully automatic manner, use the cuing paradigm as a method of investigation. It can be argued, that such tasks do not investigate the mechanisms behind cross-modal correspondences, rather investigate the effect of cross-modal correspondences on orientation of spatial attention. On the other hand, considering multisensory integration to be the underlying mechanism behind cross-modal correspondences, then factors such as temporal and spatial proximity, should be taken into consideration. Another theoretically important consideration is the proposed by Tang, Wu & Shen (2016) framework, which suggests that endogenous processes within a multisensory object can spread in both exogenous and endogenous manner. To the best of the knowledge of the author, endogenous processes behind cross-modal correspondences have not yet been investigated in terms of multisensory integration i.e. within temporal-spatial factors linked to multisensory integration. Therefore, to properly study the possible endogenous involvement into the correspondence effect, the current research uses a classic speeded discrimination paradigm, where stimuli from different modalities are presented simultaneously and for a short period of time (100ms), similar to the one used by Evans & Treisman (2010).

In addition to this, studies supporting endogenous processes (Chiou & Rich, 2012a, Chiou & Rich 2015) used strong probability manipulations formed in two modalities (particular pitch is linked to a particular spatial position of visual stimuli). Introducing such practically explicit link between the features of stimuli in different modalities, might have led to the faster activation of the associations, thus making the associations stronger, which results in faster reaction times on congruent trial than incongruent (Bugg & Crump, 2012). Moreover, participants in all of the abovementioned studies were informed of such probabilities, whether they were reverse to the natural mapping or not, which could have additionally acted beneficially and led to an increased congruency effect (Desender, 2017).

Finally, studies supporting the presence of endogenous component behind cross-modal correspondences (Chiou & Rich, 2012a, Chiou & Rich 2015) mainly use the auditory modality as task-irrelevant. According to the Colavita effect (Colavita, 1974), when presented with auditory and visual stimuli in a discrimination/detection task, participants find it harder to discriminate stimuli in auditory modality. Therefore, presenting the auditory task as task-irrelevant may also contribute to the cross-modal congruency effect in a task where the task-relevant stimuli is the visual one. In the current research, the auditory stimulus is the task relevant one, in order to increase the overall difficulty and to shift attention into a more complex auditory modality and away from a generally dominant visual modality. Moreover, auditory stimulus is used as task relevant, in order to provide replication to the results on experiments where the auditory modality was task-relevant, described by Evans & Treisman (2010).

All of these factors could have been beneficiary for the emerge of classically viewed endogenous processes linked to audio-visual cross-modal associations. This leads to the question of whether top-down influence would be visible and robust enough to manifest itself when less beneficial manipulation of time, probability and instructions introduced in behavior studies using reactions times. If the processes behind cross-modal correspondences are truly automatic, then probability manipulation should not influence the effect. If endogenous processes indeed influence the congruency advantage of cross-modal correspondences, and if the theoretical account of Tang, Wu & Shen (2016) on endogenous processes behaving in an exogenous manner is correct, then the endogenous influence should be visible in a simpler paradigm, where stimuli are presented simultaneously e.g. simple speed discrimination task similar to the one used by Evans and Treisman (2010) and Parise & Spence (2012); where manipulation of proportion is less advantageous than the one presented in the experiments conducted by Chiou & Rich (2012a,2015) and where participants are not informed for such probabilities in the instructions.

Therefore, the aim of the current research is to investigate whether endogenous processes linked to cross-modal correspondences are robust enough to manifest themselves in a paradigm without delay time between auditory and visual stimuli, manipulating the probability in only one modality and providing no explicit instructions to participants about such probabilities.

Chapter 5: Experiments

In order to investigate whether the endogenous component linked to audiovisual cross-modal correspondences is strong enough to be evident in a study using less explicit manipulations - two experiments have been conducted, using speeded discrimination paradigm. In both experiments, participants were asked to discriminate the pitch of a presented through headphones sine tone (high or low pitch), while a task-irrelevant black square appearing on screen in three positions (center, above fixation, below fixation). The first experiment presents the dimensions of the auditory and visual stimuli randomly. The second experiment introduces a probability manipulation (75% vs. 25%) in only the auditory modality i.e. a group of participants are presented with 75% high tones and 25% low tones, and another group of participants are presented with 75% low tones and 25% high tones, while presentation of visual stimuli was kept randomized. The aim of the probability manipulation is to induce endogenous attention to the task (Sarmiento, Shore, Milliken & Sanabris, 2011).

In the next sections each experiment will be discussed in detail.

5.1 Experiment 1. Replication of the cross-modal correspondence effect

5.1.1 Aims and Hypotheses

The aim of the first experiment was to investigate whether cross-modal correspondences between pitch and spatial position are present in a speeded discrimination paradigm, where the auditory stimuli is task-relevant and the visual stimuli is task-irrelevant. Thus, it aimed to replicate previous findings (Evans & Treisman, 2010), in which participants were faster to discriminate stimuli in one modality (auditory), when it was accompanied with task-irrelevant congruent stimuli in another modality (visual).

Also, the experiment aimed to serve as a control experiment. Lastly, the experiment intended

to investigate reaction times for a control condition in the visual task-irrelevant modality, rather than control unimodal condition. Most of the studies investigating cross-modal correspondences use a unimodal condition as a control (Evans & Treisman, 2010; Martino & Marks, 1999). For example, results from Evans & Treisman (2010) experiments 1 and 2 (tone vs. spatial position), showed a significant difference between congruent and incongruent trials but not between congruent and control unimodal trials. So, the introduction of a control condition in the task-irrelevant modality aims to further explore the effect the dimensions of the visual modality have on cross-modal correspondence congruency effect.

Since most of the studies use the 'congruency' type of design collapsing the 'up' and 'down' trials into congruent/incongruent and reporting no unfold results, the assumption here was that the cross-modal correspondence effect is symmetrical in the vertical space. Namely, the main hypothesis was that cross-modal correspondences between pitch and spatial position would be present and that participants would be faster to discriminate high pitch tone, when the visual stimuli presented on screen is above fixation, compared to visual stimuli presented below fixation; and low pitch tone, when the visual stimuli presented on screen are below fixation, compared to visual stimuli presented above fixation (which, when collapsed, would be reflected in congruency analysis: congruency trials would be significantly faster than incongruent trials). Based on results from Evans & Treisman (2010) study, expectations were that RT for control conditions to be similar to those of the congruent trials for all spatial positions.

5.1.2 Participants

Forty-eight people participated in the experiment (7 males, mean age 23,1). Most of them were students from New Bulgarian University who volunteered to participate in the study or in exchange for 1 course credit. Participants handedness was defined by themselves. All

participants had normal or corrected to normal vision. None of the participants reported any hearing disability.

5.1.3 Stimuli and Design

The design of the experiment was 2 (tones: high vs. low) x 3 (spatial position: center vs. down vs. up) within-subjects design (see Figure 1 for illustration of all the conditions). The dependent variable was the (RT) for discriminating the tones as either high or low pitched. Auditory stimuli included a high tone (3000Hz) and a low tone (1000Hz) created using Audible® v.2.3.2 software as sine tones with length 100ms (sample rate: 44100 Hz, bit sample 16).

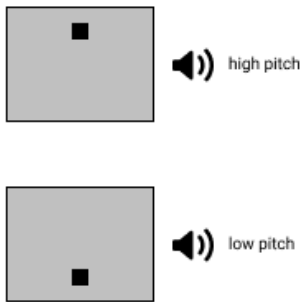
Visual stimuli included a black color square presented on a silver background. Visual stimuli (1.9° visual angle) were presented centrally, above fixation (8.7° from fixation) and below fixation (-8.7° from fixation).

5.1.4 Procedure

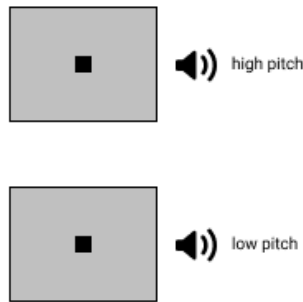
Stimuli presentation and reaction times/accuracy data gathering was done using E-prime 2.0 software (Schneider, Eschman, & Zuccolotto, 2002). Participants were positioned in individual booths in front of 22-inch monitor with resolution 1920 x 800 pixels, refresh rate 50Hz. Distance between participants and screen was ~55cm. Sound was played via headphones – with maximum volume of headphones, 70 out of 100 volume of computer. Written instructions were presented on the screen, while experimenter also read them aloud. Participants were

Type of trials

A) Congruent trials



B) Control trials



C) Incongruent trials

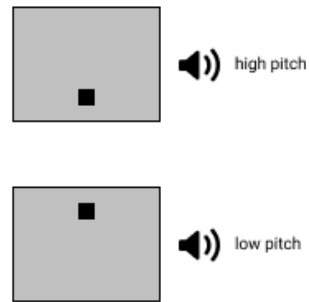


Figure 1. *Illustration of conditions. A) Congruent condition trials, B) Control condition trials, C) Incongruent condition trials.*

informed that they would hear a sound and that they would need to answer as fast and as accurate as possible whether the tone has a high pitch or a low pitch by pressing a button on the computer keyboard (“j” or “k”) with one hand. They were also asked to look at the screen at all times. The instructions didn’t include any specification that visual stimuli would also appear on screen. In the booth an experimenter was present throughout the experiment, positioned to the left of participants, pretending to read book, in order to monitor whether participants followed instructions and looked at the screen during the procedure. Response mapping was counterbalanced between subject. A trial included centrally presented fixation cross (500ms), followed by auditory and visual simultaneous stimuli presentation (100ms). After stimuli disappearance participants had 1500ms to respond. Inter-trial interval was 1000ms (see Figure 2).

Participants first went through a familiarization block of 10 trials where they heard a tone while on screen centrally appeared text specifying whether the tone, they hear is high or low pitch. After the familiarization, participants gave start to the practice set of trials by pressing a button on the computer keyboard. The practice block consisted of 12 trials with feedback after each

EXPERIMENTAL PROCEDURE

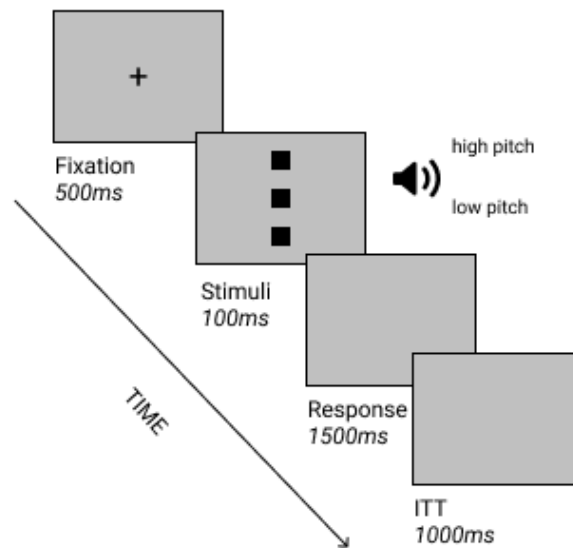


Figure 2. Trial structure. The black squares show the 3 possible position visual stimuli can appear by trial.

one. After the practice, participants gave a start to the experimental procedure by pressing a button. The experimental procedure included 168 trials. Trials in the practice and in the experimental procedure were randomized. The experiment took 10-15 minutes to finish.

5.1.5 Results and Discussion

5.1.5.1 Data Trimming

Twelve (25%) participants were removed from the data. Six (12.5%) of those were removed due to accuracy below 80%, five (10.4%) because they didn't follow instructions to look at the screen at all times and one (2%) because of technical issues. From the remaining thirty-six participants (5 male), two defined themselves as ambidexter (1 male), three as left-handed (1 male) and thirty-one as right-handed (3 male).

Only correct answers were included in the RT for data analysis. Errors (5.48%) and outliers (5.01%) defined by 2 standard deviations (SD) below and above the mean RT per subject and per condition were excluded from the RT analysis. The trimmed data were averaged by subject. To compare the results with previous studies, congruency analysis was also conducted.

5.1.5.2 Congruency - Results and Discussion

A one-way repeated ANOVA(rANOVA) was conducted with factor congruency (congruent vs. control vs. incongruent). For descriptive statistics of congruency see Table 1.

Results showed a significant main effect (see Figure 3) of congruency ($F(2,70) = 11.25, p < 0.001, \eta_p^2 = 0.24, \mu = 0.99$). A Post-hoc Bonferroni test showed that the significance comes from incongruent (356ms) condition produced slower responses than the congruent (330ms) one ($p < 0.001$) and then control (332ms) one ($p < 0.001$). There was no difference between control and congruent conditions ($p = 1.00$). This result is in agreement with previously reported unimodal control conditions (Evans & Treisman, 2010) that showed no difference between control and congruent conditions. Effect size (in ms) of congruency, was calculated by subtracting the mean RT for incongruent trials and mean RT for congruent trials and was equal to 26ms.

Table 1. Descriptive statistics for factor congruency. Mean RTs (ms), standard deviations (SD) and 95% confidence intervals (CI), averaged by subject and per condition.

Congruency condition	Mean (SD)	CI (95%)
Congruent	330 (86)	168 – 526
Control	332 (95)	166 – 510
Incongruent	356 (88)	196 – 536

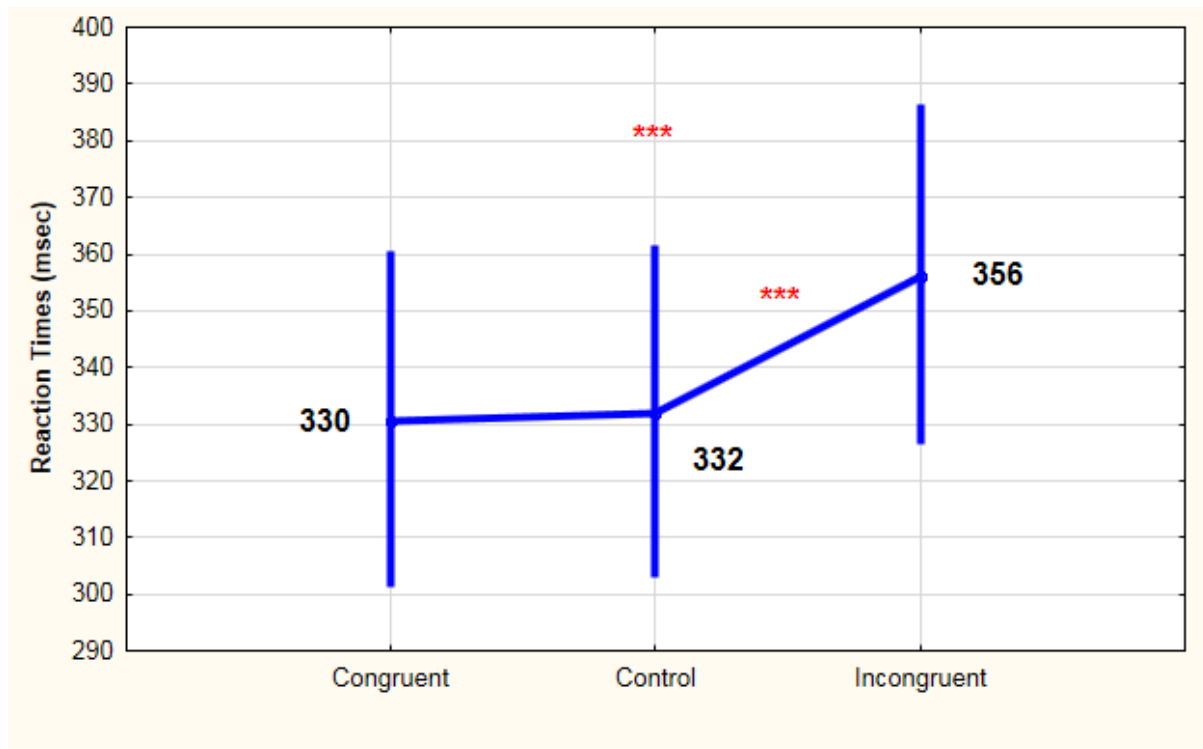


Figure 3. Main effect of congruency. Vertical bars denote standard errors. * $p < .05$, ** $p < .01$, *** $p < .001$

Overall, the results supported previous findings (Evans & Treisman, 2010) and showed a cross-modal correspondence congruency effect between pitch and spatial position in a speeded-discrimination paradigm, where the auditory stimulus was task-relevant and the visual stimulus was task-irrelevant. People responded faster to tones in congruent condition than in incongruent. Additionally, the analysis showed the difference between incongruent and control/congruent conditions and not between control and congruent condition, which suggest that the cross-modal correspondence effect is a result of an interference, rather than facilitation.

Next analysis intended to unfold the congruency and better understand the dynamics behind cross-modal correspondences and whether the effect was present across all conditions on tone and position. As mentioned above, this was done because most of the studies examining cross-modal correspondences reported generally results for congruent vs. incongruent trials.

5.1.5.2 Tone vs. Position– results and discussion

A two-way rANOVA subject analysis was conducted with factors tone (high vs. low) and spatial position (center vs. down vs. up). Table 2 presents descriptive statistics per condition.

rANOVA obtained a significant main effect of pitch ($F(1,35) = 4.92$, $p = 0.03$, $\eta_p^2 = 0.12$, $\mu = 0.57$), suggesting that participants judged high tones (332ms) faster than low tones (348ms). Main effect of spatial position was also significant ($F(2,70) = 3.98$, $p = 0.02$, $\eta_p^2 = 0.10$, $\mu = 0.69$). A Post-hoc Bonferroni analysis showed that the significance comes from the center (323ms) position produced faster responses than the down (362ms) position ($p = 0.036$). There was no difference between control and congruent conditions ($p = 0.074$).

Lastly, rANOVA analysis showed a significant two-way interaction between tone and position ($F(2,70) = 9.29$, $p < 0.001$, $\eta_p^2 = 0.20$, $\mu = 0.97$). Figure 4 presents the interaction. A Post-hoc comparison using Bonferroni showed that participants were significantly faster to respond to high tones (mean = 343ms), compared to low tones (mean = 380ms) when the stimuli presented on the screen were above fixation ($p < 0.001$), suggesting significant cross-modal correspondence effect only for pitch and visual stimuli presented in upper position.

Table 2. Descriptive statistics per condition. Mean RTs (ms), standard deviations (SD) and 95% confidence intervals (CI), averaged by subject and per condition.

Spatial Position	High Tone		Low Tone	
	Mean (SD)	CI (95%)	Mean (SD)	CI (95%)
Center	322 (84)	294 – 350	342 (93)	310 – 374
Down	350 (90)	319 – 380	338 (101)	330 – 393
Up	323 (80)	296 – 380	362 (94)	331 – 394

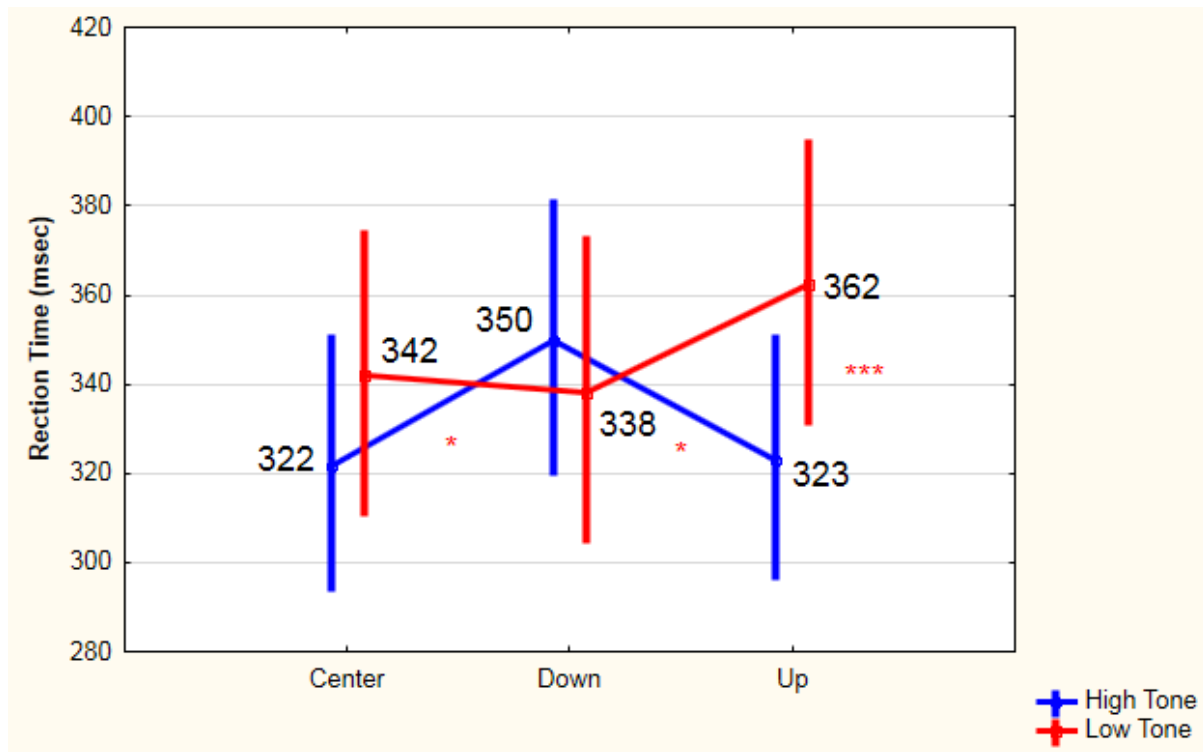


Figure 4. Interaction between tone and spatial position. Vertical bars denote standard errors. * $p < .05$, ** $p < .01$, *** $p < .001$

Significance was also found between high tones/center position (322ms) and low tones/ up position (362ms) ($p < 0.001$); high tones/ down position (350ms) and high tones/ up position (323ms) ($p = 0.03$); high tones/ central position (322ms) and high tones/ down position (350ms) ($p = 0.02$).

The results from the tone and position rANOVA analysis showed that cross-modal correspondences effect was present only for pitch and visual stimuli presented in upper position.

In summary, results from Experiment 1 revealed cross-modal correspondence between pitch and spatial position in the context of the presented task – speeded discrimination of a task-relevant auditory stimuli, while simultaneously task-irrelevant visual stimuli were presented

on a computer screen. Participants in the experiment were faster to respond to congruent and control trials, compared to incongruent trials, revealing that cross-modal correspondences are the results of an interference between the task-irrelevant feature of stimuli in one modality and the task-relevant feature of stimuli in another modality. Lastly, an analysis looking into the particular dimensions of the associations, showed that in fact, only the upper visual position was responsible for the congruency effect. The lower visual position showed no effect.

One explanation behind this result can be drawn from studies, investigating asymmetries in information processing between upper visual field (UVF) and lower visual field (LVF) (Previc, 1990, Previc & Blume, 1993, Zhou, Yu, Yu, Wu and Zhang, 2017). It has been suggested that stimuli are processed differently depending on whether they occur in the UVF or in the LVF, and that these asymmetries are determined on the distance between the subject and the object and which visual stream of processing is used (Previc, 1990). The visual distance can be determined as peripersonal (an object reachable by the subject) and extrapersonal (an out-of-reach item) space. Tasks performed in extrapersonal space show an UVF advantage in discrimination tasks measuring RT and accuracy, while peripersonal space show an LVF advantage (Previc & Blume, 1993, see Johansson, Tagu & Kristjansson, 2018). Visual stream of processing refers to the two streams of visual information processing – the ventral and the dorsal pathways. LVF has stronger connections with the dorsal pathway, which is linked with visually guided actions, while the UVF is more connected to the ventral stream, which is associated with perceptual identification of objects (Goodale & Milner, 1992). Therefore, tasks involving actions show an advantage in the LVF and tasks involving recognition/detection show an UVF advantage (Previc & Blume, 1993, see Johansson, Tagu & Kristjansson, 2018). The advantage of UVF in tasks involving detection, can explain why cross-modal correspondence effect was present only for the up spatial position i.e. participants were faster

to detect and process stimuli presented in the UVF, which could have resulted in quicker formation of cross-modal associations. Thus, when presented with low tones, participants found it harder to ignore the stimuli presented in upper position. Moreover, the UVF advantage could also explain the lack of significance for low tones and down spatial position – squares presented below fixation were slower to detect, leading to a weaker association between spatial position and tone pitch.

The next experiment intends to further investigate cross-modal correspondences by introducing a manipulation aimed to induce endogenous processes and by comparing these results to the congruency and tone/position data of Experiment 1.

5.2 Experiment 2 – Pitch probability manipulation

5.2.1 Aims and Hypotheses

The aim of the second experiment was to investigate whether the endogenous component presumably linked to cross-modal correspondences (Getz & Kubovy, 2018; see Spence & Deroy, 2013) is strong enough to be visible in a study that uses less explicit manipulations to induce endogenous processes. Recall that in accordance with temporal factors defining multisensory integration and the framework proposed by Tang, Wu & Shen (2016), the endogenous component should be visible in a paradigm presenting stimuli simultaneously for a 100ms. In order to test it, Experiment 2 prompts endogenous processes by introducing probability manipulation (Sarmiento, Shore, Milliken & Sanabris, 2011) in auditory modality only. One group of participants were introduced to the High tone probability (75% of trials included a high tone, 25% of trials included low tone), while another group of participants was introduced to the Low tone probability (75% of trials included a low tone, 25% of trials – high

tone). Unlike most of the studies, here the participants remained explicitly uninformed about the manipulation.

The main hypothesis was that if endogenous component is strong enough then results would show a change (increase or decrease) in the congruency effect size compared to the control condition, due to the probability manipulation. Consequently, another hypothesis was that results for factors tone vs. spatial position should also show a change (increase or decrease) in the cross-modal correspondence effect between high and low tones, for each of the two probability groups. A change of both congruency effect size and cross-modal correspondence effect would suggest that the manipulation has managed to affect participants performance on the task, which would suggest that endogenous processes could be involved in the cross-modal correspondence effect and that it could be traced in a paradigm using less explicit manipulation.

5.2.2 Participants

Ninety-six people participated in the experiment (26 males, mean age 23,9). Most of them were students from New Bulgarian University who volunteered to participate in the study or in exchange for 1 course credit. Participants handedness was defined by themselves. All participants had normal or corrected to normal vision. None of the participants reported any hearing disability. Ten of the participants which conducted Experiment 2 also conducted Experiment 1, with the closest time period between the two experiments to be 2 days and the farther period – around 3 months.

5.2.3 Stimuli and Design

The design of the experiment was 2 (tones: high vs. low) x 3 (spatial position: center vs. down vs. up) x 2 probability (75% vs. 25%) within subjects' design.

The dependent variable was the reaction time (RT) for discriminating the pitch.

Auditory and visual stimuli were the exact same as Experiment 1.

5.2.4 Procedure

The procedure setting was the same as Experiment 1.

The differences were the following. Participants were randomly assigned to a group with high tone probability of 75% or low tone probability of 75%. No information about the probability was given to participants. The same experimenter was present in the booth, monitoring whether participants followed instructions. Participants went through experimental procedure consisting of 216 trials. After the 108th trial, a break screen was presented, informing subjects that in order to continue with the procedure they need to press a button on the keyboard. Participants choose how long should the break be. The experiment took 10-15 minutes to finish.

5.2.5 Results and Discussion

5.2.5.1 Data Trimming

Twenty-four (25%) participants were removed from the data. Thirteen (13.5%) of those were removed due to accuracy below 80% (11 from 75% high tone probability group), eleven (11.5%) because they didn't follow instructions to look at the screen at all times (7 from 75% high tone probability group). From the remaining seventy-two participants (19 male), one defined themselves as ambidexter (1 female), four as left-handed (1 male) and sixty-seven as right-handed (18 male). Thirty-seven participants were in the 75% high tone probability group, thirty-five participants in the 75% low tone probability group.

Only correct answers were considered for RT data analysis. Errors (4.90%) and outliers (4.85%) defined by 2 standard deviations (SD) below and above the mean RT per subject and per condition were excluded from the RT analysis. The trimmed data were averaged by subject. To compare the results with Experiment 1, effect size analysis was also conducted. All post-hoc analysis were done using Bonferroni.

5.2.5.2 Congruency Analysis – results and discussion

A mixed rANOVA was conducted with factor congruency (congruent vs. control vs. incongruent) a within-subject and group (75%High pitch vs. 75%Low pitch) as a between subject factor. For descriptive statistics of congruency see Table 3. Results showed a significant main effect of congruency ($F(2,140) = 31.98, p < 0.001, \eta_p^2 = 0.31, \mu = 1.00$). Post-hoc analysis showed that significance comes from incongruent (311ms) condition, which produced slower responses than congruent (293ms) one ($p < 0.001$) and control (289ms) one ($p < 0.001$). A significant main effect of pitch probability ($F(1,70) = 4.05, p = 0.048, \eta_p^2 = 0.05, \mu = 0.51$) was also found, suggesting that performance in the two between subject groups was different.

Finally, rANOVA showed a significant interaction (see Figure 5) between congruency and group ($F(2,140) = 3.66, p = 0.028, \eta_p^2 = 0.05, \mu = 0.67$). Post-hoc analysis showed that participants were significantly faster to respond to control trials (271ms) when presented with mostly high tones, compared to incongruent trials (336ms) when presented with mostly low tones ($p = 0.033$). In the 75%High pitch group, the only significant difference was between incongruent (286ms) trials and control (271ms) ones ($p = 0.003$). In the 75%Low pitch group participants were significantly slower to respond to incongruent trials (336ms) compared to

Table 3. Descriptive statistics for factors congruency and group – Experiment 2. Mean RTs (ms), standard deviations (SD) and 95% confidence intervals (CI), averaged by subject and per condition.

Congruency condition	75% High Tone		75% Low Tone		Total Mean (SD)
	Mean (SD)	CI (95%)	Mean (SD)	CI (95%)	
Congruent	276 (89)	246 – 305	311 (85)	282 – 340	293 (89)
Control	271 (85)	242 – 299	308 (86)	279 – 338	289 (87)
Incongruent	286 (87)	257 – 315	336 (94)	304 – 369	310 (94)

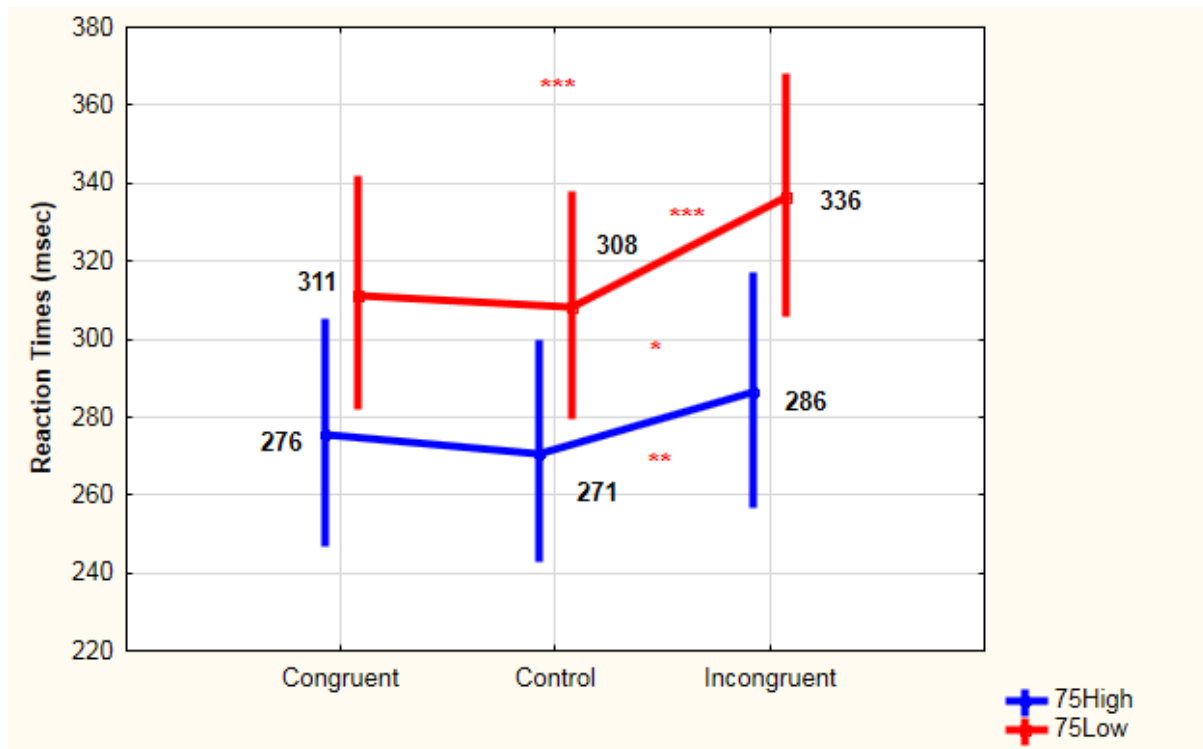


Figure 5. Experiment 2 –interaction between congruency (congruent vs. control vs. incongruent) and pitch probability (75High vs. 75low) mixed ANOVA analysis. Vertical bars denote standard errors. * $p < .05$, ** $p < .01$, *** $p < .001$

control (308ms) trials ($p < 0.001$) and congruent (311ms) trials ($p < 0.001$).

5.2.5. Effect size comparison

In addition, a mixed rANOVA was conducted with position (down and up) as within-subject factor and experiment (Experiment 1 (50%/50%), 75%High pitch and 75%Low pitch) as between subject factor. For descriptive statistics of congruency see Table 4. The dependent variable was effect size (in ms). Mean effect size (in ms), was calculated by subtracting the mean Incongruent RT from the mean Congruent RT per subject and per spatial position. Thus, positive effect size suggests faster performance on congruent trials, and negative effect size – faster performance on incongruent trials.

Table 4. Descriptive statistics per condition. Mean effect size in ms (Incongruent Rt – Congruent RT), standard deviations (SD) and 95% confidence intervals (CI), averaged by subject and per condition.

Spatial Position	High Tone		Low Tone	
	Mean (SD)	CI (95%)	Mean (SD)	CI (95%)
Experiment 1 (50%/50%)	11 (73)	-13 – 36	39 (52)	21 - 57
Experiment 2 (75%High pitch)	-34 (46)	-49 – -18	62 (74)	37 - 86
Experiment 2 (75%Low pitch)	17 (47)	1 - 33	26 (55)	7 - 45

The analysis showed a significant main effect of position ($F(1,105) = 23.58, p < 0.001, \eta_p^2 = 0.18, \mu = 0.99$). Participants were faster to respond to congruent trials in the upper position (42.51ms), and faster to respond to incongruent trials in the down position (-1.79ms). There was no significant main effect of experiment ($F(2,105) = 0.98, p = 0.376, \eta_p^2 = 0.01, \mu = 0.21$). rANOVA analysis also showed a significant two-way interaction between position and experiment ($F(2,105) = 8.50, p < 0.001, \eta_p^2 = 0.12, \mu = 0.96$). Figure 6 presents the interaction. Post-hoc analysis showed participants in the 75%High pitch experiment were significantly faster to respond to incongruent trials presented in lower spatial position, resulting in negative effect size (-34ms), compared to faster responses to congruent trials (11ms) in lower spatial position in Experiment 1 ($p < 0.001$); and faster responses to congruent trials (17ms) in 75%Low pitch experiment ($p = 0.004$). Significant difference was also found between effect sizes in 75%High pitch experiments, where participants were significantly faster to respond to congruent trials (62ms) and faster to respond to incongruent trials (-34ms) in down position ($p < 0.001$).

Overall, results from the congruency analysis from Experiment 2 showed the probability manipulation was successful in Experiment 2 – 75%High pitch group. The negative effect size in the down spatial positions, means that participants responded faster to incongruent trials,

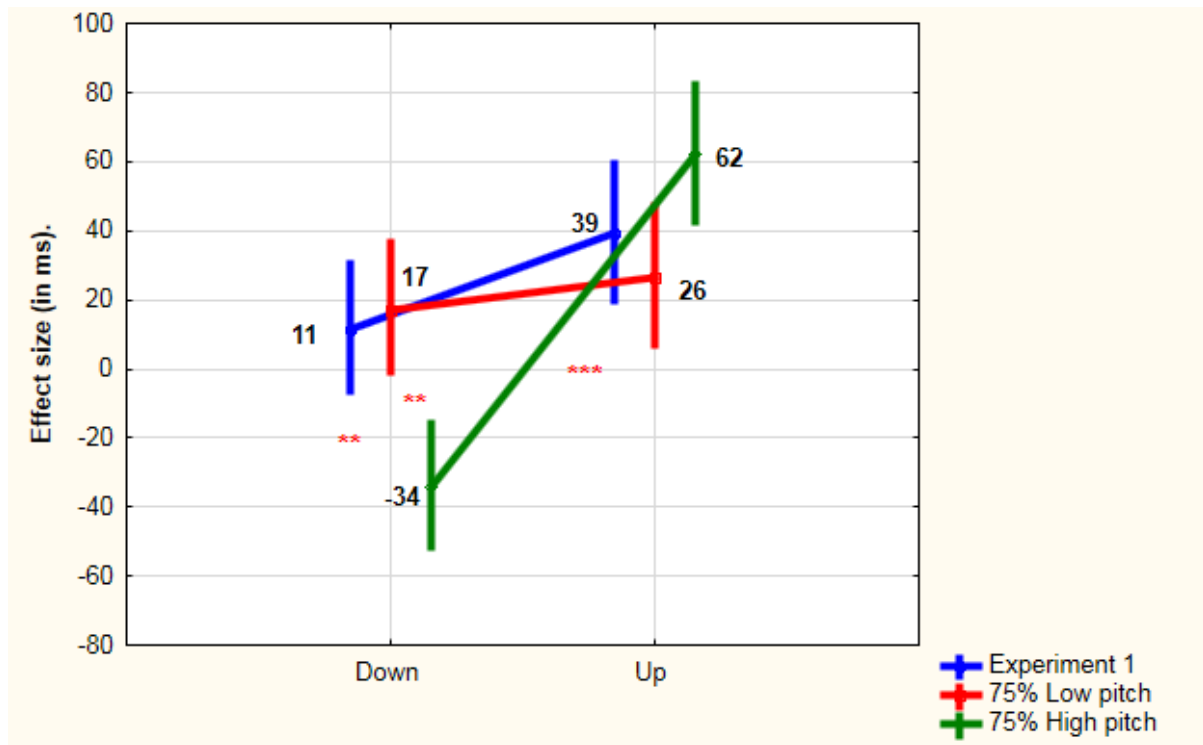


Figure 6. Interaction between position (down vs. up) and experiment (Experiment 1 vs. 75%High pitch vs.75%Low pitch). Dependent variable effect size (ms) calculated by subtracting from mean Incongruent RT - mean Congruent RT per subject per condition. Vertical bars denote standard errors. * $p < .05$, ** $p < .01$, *** $p < .001$

hence eliminating the congruency effect between pitch and spatial position. On the other hand, results from 75%Low pitch group showed similar results to Experiment 1, implying that probability manipulation was not able to influence congruency advantage for low tone.

An additional ANOVA was conducted to further explore whether and how the features of tone and position have been affected by the probability manipulation in the two pitch probability groups.

5.2.5.4 Tone vs. Position vs. Pitch probability - results and discussion

Repeated measures ANOVA subject analysis with tone (high vs. low), spatial position (center vs. down vs. up) as within-subject factors and pitch probability (75High vs 75Low) as between-subjects factor was conducted. For descriptive statistics see Table 5. The analysis showed a

Table 5. Descriptive statistics for factors tone, spatial position and group – Experiment 2. Mean RTs (ms), standard deviations (SD) and 95% confidence intervals (CI), averaged by subject and per condition.

Condition (Tone/Spatial position)	75% High Tone		75% Low Tone		Total Mean (SD)
	Mean (SD)	CI (95%)	Mean (SD)	CI (95%)	
High/Center	259 (89)	229 – 289	332 (94)	300 – 364	294 (98)
High/Down	275 (86)	246 – 304	328 (81)	300 – 356	301 (87)
High/Up	265 (90)	235 – 295	313 (71)	287 – 338	288 (85)
Low/Center	311 (80)	285 – 338	301 (87)	271 – 331	306 (83)
Low/Down	309 (100)	276 – 342	311 (91)	280 – 342	310 (95)
Low/Up	327 (104)	293 – 305	339 (100)	305 – 374	330 (102)

main effect of tone ($F(1,70) = 20.73$, $p < 0.001$, $\eta_p^2 = 0.23$, $\mu = 0.99$) and main effect of spatial position ($F(2,140) = 6.26$, $p = 0.002$, $\eta_p^2 = 0.08$, $\mu = 0.89$). Post-hoc analysis displayed the significance to be between the center and up position ($p = 0.002$), with people responding faster in trials where the square was presented centrally (mean = 301ms) compared to above fixation (mean = 311ms). The ANOVA analysis also showed a two-way interaction between tone and experiment position ($F(1,70) = 37.64$, $p < 0.001$, $\eta_p^2 = 0.35$, $\mu = 0.99$). Post-hoc Bonferroni showed that the significant main effect is a result of the difference between high and low tones in the 75High group ($p < 0.001$) and between high tones in the 75High group and high tones in the 75Low group ($p = 0.036$). No other significant effects were found for the tone/experiment two-way interaction (all $ps > 0.9$).

The analysis also showed significant two-way interaction between tone and position ($F(2,140) = 12.651$, $p < 0.001$, $\eta_p^2 = 0.15$, $\mu = 0.99$). Post hoc Bonferroni analysis showed significance between high and low tones in the up spatial position ($p < 0.001$), suggesting a significant cross-modal correspondence effect between pitch and stimuli presented above fixation. Other significant results included: High tones/Center position and Low tone/ Up position ($p < 0.001$);

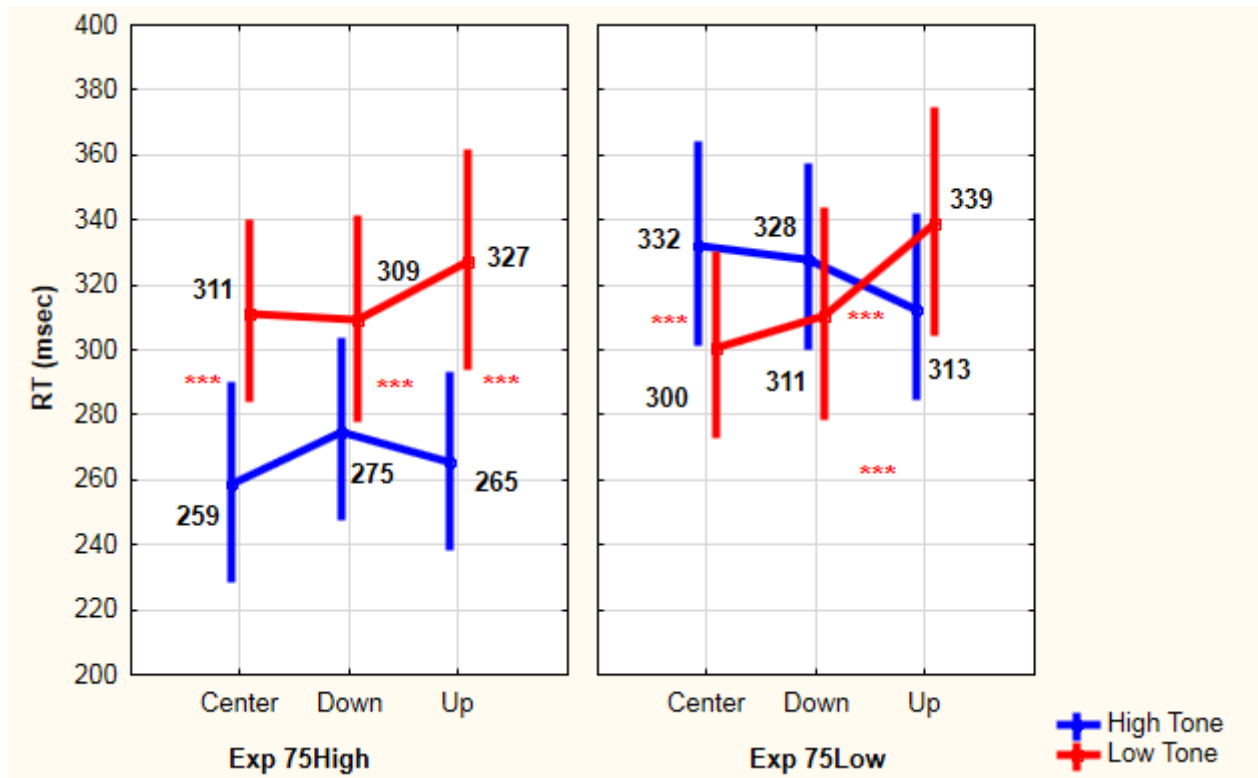


Figure 6. Experiment 2 –three-way interaction between tone, spatial position and group. Vertical bars denote standard errors. * $p < .05$, ** $p < .01$, *** $p < .001$

High tone/Down position and Low tone/Up position ($p < 0.001$); High tone/ Up position and Low tone/Center position ($p = 0.027$); High tone/Up position and Low tone/Down position ($p = 0.002$); Low tone/Center position and Low tone/Up position ($p < 0.001$) and Low tone/Down position and Low tone/Up position ($p < 0.001$).

Lastly, analysis showed a three-way interaction between tone, spatial position and experiment ($F(2,140) = 4.74$, $p = 0.010$, $\eta_p^2 = 0.06$, $\mu = 0.78$). See Figure 6 for a visual representation of the interaction. Post-hoc analysis showed a significant difference between high and low tones in all position – central ($p < 0.001$), down ($p < 0.001$) and up ($p < 0.001$). In the 75Low group significant difference was found between High tones/ Central position and Low tones/Central position ($p = 0.11$); Low tones/ Center position and Low/Tones Up position ($p < 0.001$) and Low tones/ Down position and Low/Tones Up position ($p = 0.042$).

The results further confirmed the congruency analysis and showed that the probability manipulation used in Experiment 2, led to a change in participants performance. Nevertheless, this change was different in the 75%High pitch group and in the 75%Low pitch group. In the 75%High pitch group, participants responded faster to high tones, despite the location of the visual stimuli presented on the screen i.e. responses to high tones were faster even on incongruent trials. This suggests, that the probability manipulation was able to change the cross-modal correspondence effect, showing that the effect is not immune to endogenous processes. On the other hand, results from the 75%Low pitch group, didn't show such evident change due to the probability manipulation. Participants in the 75%Low pitch group found it easier to respond to High tones in upper spatial position despite the low probability, similar to results from Experiment 1, implying that the probability manipulation was not able to influence associations between high tone and upper spatial location. Nevertheless, results in the control condition, did show manipulation effect – participants were significantly faster to respond to low tones in control condition, compared to Experiment 1. So, the higher presentation of low tones managed to increase the cross-modal associations between low tones and centrally presented visual stimuli.

In summary, results from Experiment 2, showed that probability manipulation did lead to a change in performance, but this change was different across the pitch probability groups. In order to investigate further endogenous processes and trace the dynamics of it distribution analysis was conducted.

5.3 Distribution analysis

A distribution analysis was conducted for RTs in order to see how long does it take for the cross-modal correspondence effect to emerge. Moreover, the distribution analysis aims to provide an insight into exogenous and endogenous processes behind cross-modal correspondences found in Experiment 1 and 2, from the current research. More specifically, distribution analysis can specify how did the probability manipulation change performance by specifying the differences in RTs across all bins.

A distribution analysis with RTs from Experiments 1 and 2 were rank-ordered for tone and spatial condition. Results from each condition was divided into five bins (20% in each bin). Mean RTs per subject and per condition were computed for each bin. A rANOVA within subject's analysis with factors tone (high vs. low), spatial position (center vs. down vs. up) and bin (1-5) was conducted per experiment. Post-hoc analysis was performed using Bonferroni. Results from Experiment 1 showed a significant main effect of position ($F(2,70) = 4.03$, $p = 0.021$, $\eta_p^2 = 0.10$, $\mu = 0.70$). Participants were faster on trials, in which the visual stimuli were centrally presented (330ms) compared to trials in which visual stimuli were in lower (342ms) spatial position ($p = 0.038$). rANOVA also showed significant main effect of tone ($F(1,35) = 4.51$, $p = 0.040$, $\eta_p^2 = 0.11$, $\mu = 0.54$). Participants were faster to respond to high pitched tones (330ms) compared to low pitched tones (346ms). Significant main effect of bins ($F(1,140) = 333.92$, $p = 0.001$, $\eta_p^2 = 0.21$, $\mu = 0.97$) was also found). Post-hoc analysis showed that significant difference was present across all bins (all $ps < 0.001$). Lastly, significant two-way interaction was found between position and tone ($F(2,70) = 9.74$, $p < 0.001$, $\eta_p^2 = 0.21$, $\mu = 0.97$).

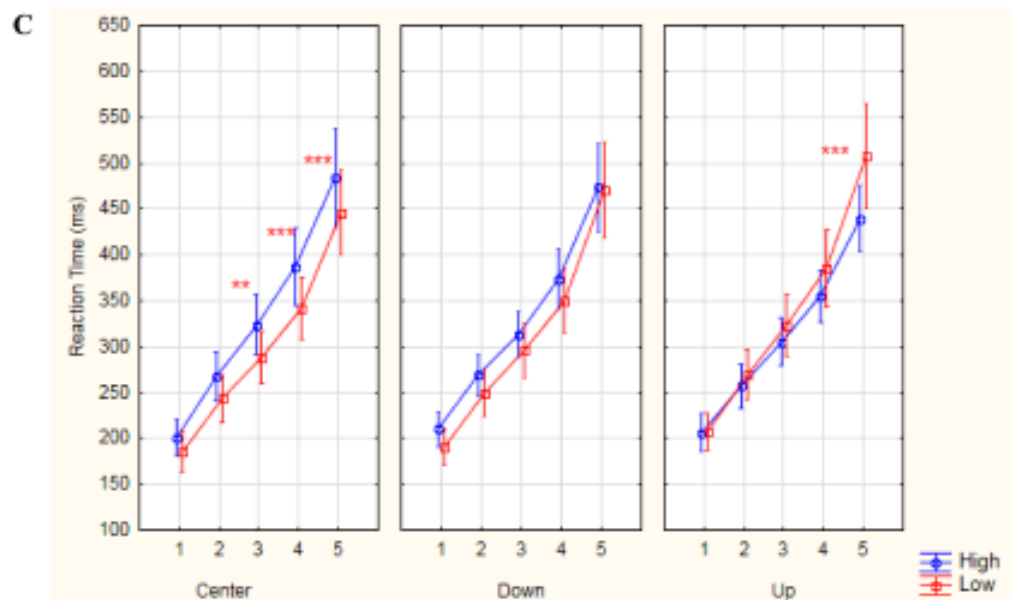
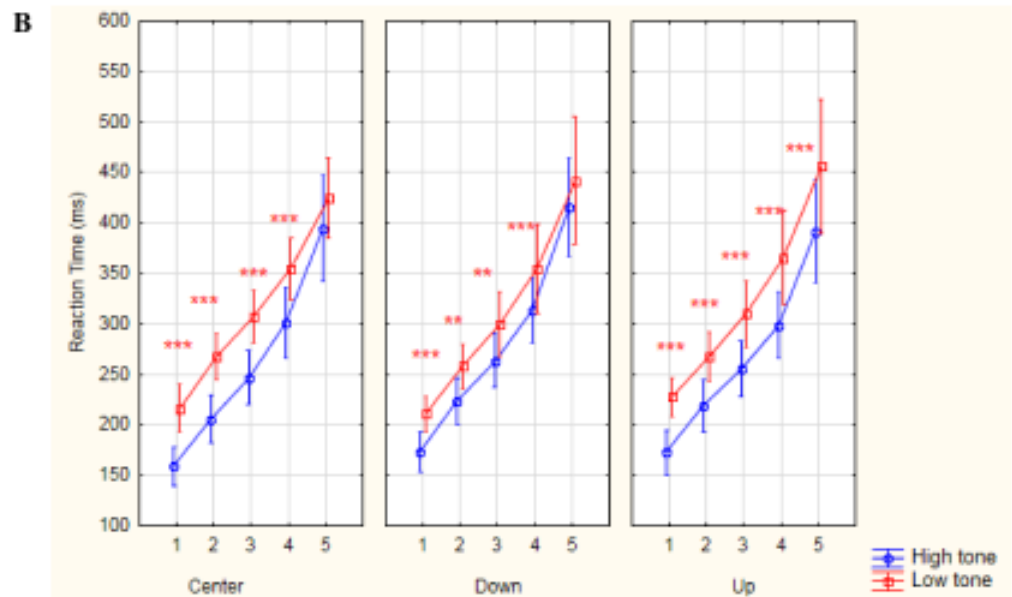
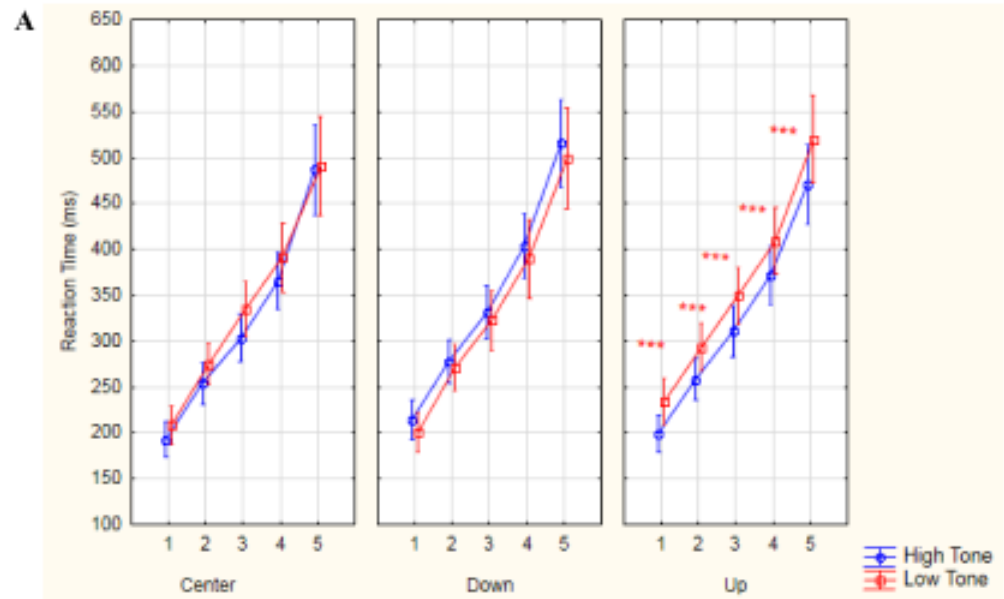
Results from Experiment 2 - 75%High pitch group showed a significant main effect of position ($F(2,70) = 3.40$, $p = 0.038$, $\eta_p^2 = 0.08$, $\mu = 0.63$). Participants were faster on trials, in which the

visual stimuli were centrally presented (287ms) compared to trials in which visual stimuli were in lower (295ms) spatial position or upper spatial position (296ms), although the post-hoc analysis didn't show any significance between spatial positions. rANOVA also showed significant main effect of tone ($F(1,35) = 50.76, p < 0.001, \eta_p^2 = 0.59, \mu = 1.00$). Participants were faster to respond to high pitched tones (269ms) compared to low pitched tones (317ms). Significant main effect of bins ($F(1,140) = 135.20, p = 0.001, \eta_p^2 = 0.79, \mu = 1.00$) was also found. Post-hoc analysis showed that significant difference was present across all bins (all p s < 0.001). No other significant main effects or interactions were found.

Results from Experiment 3 - 75%Low pitch group showed a significant main effect of bins ($F(4,136) = 277.83, p < 0.001, \eta_p^2 = 0.89, \mu = 1.00$). Post-hoc analysis showed that significant difference was present across all bins (all p s < 0.001). rANOVA also showed significant two-way interaction between position and tone ($F(2,68) = 15.89, p < 0.001, \eta_p^2 = 0.31, \mu = 0.99$). Also, significant two-way interaction was found between tone and bins ($F(4,136) = 5.59, p < 0.001, \eta_p^2 = 0.11, \mu = 0.94$). Lastly, significant three-way interaction was found between position, tone and bins ($F(8,272) = 4.74, p < 0.001, \eta_p^2 = 0.12, \mu = 0.99$).

Taken together, RT distribution analysis of Experiments 1 and 2 show an obvious difference of cross-modal correspondence effect prompted by the probability manipulation. For a graphic representation of results see Figure 7. In Experiment 1, cross-modal correspondence effect was visible only for high pitched tones in up spatial conditions in all bins. No significant effect was present for low tones in down condition across bins. In Experiment 2 - 75%High group, there was a constant effect of the probability manipulation across all spatial conditions and bins. Consequently, the effects of endogenous processing are visible from the fastest to the slowest bins, suggesting that the endogenous manipulation was powerful and it penetrated the cross-

Figure 7. RTs for tones and spatial position, plotted for bins 1 through 5 and per experiment. A) Experiment 1, B) Experiment 2 - 75%High pitch group, C) Experiment 2 - 75%Low pitch group. x-axis present mean RT for each of the bins per position. Blue line represents high tones and red line represents low tones. * $p < .05$, ** $p < .01$, *** $p < .001$.



modal correspondence effect in an exogenous manner. On the other hand, in the 75%Low pitch group, the probability manipulation was not dominant – it was only visible in the control condition. Furthermore, the effect for low pitched tones presented in central position, developed in time. Similarly, and despite the low probability of high tones, a cross-modal correspondence effect was visible in the up spatial position, but only in the last stage. Based on these results it can be argued that the endogenous component behind cross-modal correspondences is visible in the 75%Low pitch group.

In summary, the distribution analysis of RTs across the 2 experiments, supported the theoretical account of Tang, Wu & Shen (2006). Results implied that endogenous processes are involved in cross-modal correspondences, and they can act in an exogenous manner (Experiment 2 – 75%High pitch group) and in an endogenous manner (Experiment 2 – 75%Low pitch group). Analogous results can be seen in a distribution analysis of effect size for Experiment 1 and 2 (see Appendix 1).

Chapter 6: General Discussion

In the last couple of decades, processes behind audiovisual cross-modal correspondences have been the topic of an ongoing debate in the literature (see Spence & Deroy, 2013). One of the main question researchers explore is whether cross-modal correspondences occur in an automatic manner, or whether there is an endogenous component present.

While some studies have suggested that cross-modal correspondences occur in an automatic manner (Evans & Treisman, 2010; Parise & Spence, 2012; Mossbridge, Grabowecky & Suzuki, 2011), others have suggested that there is an endogenous component involved in processes behind cross-modal correspondence (Chiou & Rich, 2012a, Chiou & Rich, 2015).

Nevertheless, these studies use rather diverse methodologies, which has led to a disparity in the research on the topic. The current study aimed to define these disparities and to address them methodologically.

In order to do so two experiments have been conducted. The first experiment aimed to show that cross-modal correspondences between pitch and position were present in the context of the paradigm used. Moreover, Experiment 1 was to serve as a control experiment. Data was analyzed for both congruency effects and for cross-modal correspondence effects within the mapping of tone and position. Particularly, to investigate whether there are differences any correspondence asymmetries across the vertical positions.

Results from Experiment 1 replicated previous findings (Evans & Treisman, 2010) and showed participants responded slower to incongruent trials, compared to congruent and control trials. Importantly, RTs for congruent and control trials did not differ significantly, suggesting that congruency advantage was due to an interference rather than facilitation. Experiment 1 also showed that cross-modal effects within the mapping of tone and position were present only for tones presented in upper spatial position i.e. participants were significantly faster to respond to

high pitched tones when visual stimuli presented on screen was above fixation, rather than below fixation.

One explanation behind such results can be drawn from studies investigating vertical asymmetries in visual information processing (Previc, 1990). It has been suggested that stimuli are processed differently depending on whether they occur in the UVF or in the LVF. Specifically, LVF has stronger connections with the dorsal pathway, which is linked with visually guided actions, while the UVF is more connected to the ventral pathway, which is associated with perceptual identification of objects (Goodale & Milner, 1992). Consequently, there is LVF advantage in tasks involving motion and coordinate spatial judgments, while UVF advantage is present for visual search and categorical judgment tasks (Thomas & Elias, 2011; Johannesson, Tagu & Kristjansson, 2018). In addition, a study using eye-tracker showed a stronger UVF bias when stimuli presentation was 150ms, while LVF bias was present during longer presentation time (10s). Considering the paradigm used in the current research, it could be that squares presented above fixation were processed faster, due to the task and the short presentation of stimuli. A bias towards the UVF resulted in faster detection of squares presented above fixation, which could have led to stronger associations between tone and position. The advantage of UVF, could also account for visual stimuli presented below fixation – it could be that stimuli were not processed as fast, meaning they needed more time to form cross-modal correspondences. Thus, weaker the cross-modal correspondence effect was weaker, as it can be seen by the lack of significance between low tones and visual stimuli presented below fixation.

The second experiment aimed to investigate the endogenous component linked to cross-modal correspondences by introducing simple manipulation to induce endogenous processes. By doing so, the experiment addresses a number of factors defined by the previously mentioned

disparity in the research on the topic. Most importantly, the experimental procedure presents stimuli simultaneously for a short period of time, similarly to the experiments conducted by Evans & Treisman (2010) and in accordance to the established time-space conditions of occurrence of multisensory integration (Talsma, Senkowski, Soto-Faraco, & Woldorff, 2010). To the best of the knowledge to the author, such methodology has not been used so far to investigate endogenous processes involved in cross-modal correspondences. Secondly, the experiment introduced a simple probability manipulation as a mean to induce endogenous processes without mentioning the existence of such manipulation in the instructions given to participants. This is in contrast to, with other studies described in detail in previous chapters (Chious & Rich, 2012a; Klapetek et al, 2012). The main assumption was that the endogenous component would be strong enough to manifest itself in the context of the procedure used in the current research. This hypothesis was mainly based on a theoretical account for interactions of multisensory integration with endogenous and exogenous attention, proposed by Tang, Wu & Shen (2016). The authors make several suggestions, but of interest to the current research is the assumption that “within a multisensory object, endogenous attention can spread from one modality to another in an exogenous manner “(Tang, Wu & Shen, 2016, p.208). Since, Experiment 2 presents stimuli simultaneously it could be argued that multisensory integration would be the underlying mechanism behind cross-modal correspondences. So, if indeed endogenous attention can spread in an exogenous manner, then results from Experiment 2 should show a change of congruency effects and a change in RTs within the polarities of cross-modal mapping, compared to Experiment 1.

Results from Experiment 2 partially supported this notion. Differences in congruency and cross-modal correspondence effect in both probability groups (75% high pitched tones group/ 75% low pitched tones group) compared to Experiment 1 did occur. However, the probability manipulation was more successful in the 75%High pitch group. Results showed that the

probability manipulation was managed to penetrate both congruency and the cross-modal correspondences in the 75%High pitch group. Specifically, participants were faster to respond to high tones, regardless of the position of square presented on screen.

In contrast, results from the 75% Low pitch group, showed that the probability manipulation was not able to influence the cross-modal correspondence effect, except in the control condition (participants responded significantly faster to low tones, compared to high tones in the control condition). Importantly, the manipulation was not as dominant as in the 75%High pitch group, and was not able to influence the cross-modal association between high pitch and upper spatial position, despite the smaller presentation of high tones. As mention previously, this difference between high- and low-pitched tones could be attributed to asymmetries of visual field processing.

Finally, a distribution analysis was conducted that aimed to investigate how did the probability manipulation change performance across 5 bins.

Results from Experiment 1, showed that the cross-modal correspondences between high pitch and upper spatial position was present and constant across all bins.

In the 75%High pitch group, significant difference in RT between high pitch and low pitch tones was present in all spatial conditions and across all bins. Therefore, the manipulation was penetrated the cross-modal correspondences effect, suggesting that processes behind cross-modal correspondences are not fully automatic. If the processes were fully automatic, then the simple manipulation should not manage to change RTs. However, since the effects of endogenous processing were visible even in the fastest bins, it could be said that they affected performance in an exogenous manner.

In the 75%Low pitch group, the significant difference in RT between high and low pitch in the control condition, developed in time. Additionally, significant difference was found in the up spatial position but only in the last bin. So, the distribution analysis show that cross-modal

correspondences needed time to emerge, which is a clear indication of endogenous processes. Taken together, results from Experiment 1 and 2, show that endogenous processes are involved in cross-modal correspondences and that they can act in both exogenous and endogenous manner.

In conclusion, a couple of limitations and weaknesses of the current research should be considered. One weakness of the experiment is the requirement that participants need to look at the screen at all times. Although an experimenter was present in the room monitoring whether participants followed the requirement, it is not enough to maintain strong control over participants performance. A possible solution for this is in next experiments, participants to be asked to lay they head on a chin rest and to track eye saccades with eye-tracker. Another limitation to the experiment is that no information was gathered of participants music experience. For example, participants which had music education, could have had stronger associations between the pitch of the tone and the response button i.e. the left key is linked with low tones and right key is linked to high tones. The compatible or incompatible response mapping could have affected performance on the task. So, future research could investigate how musicians and non-musicians perform the task, and whether the same differences seen in the 75High and 75Low condition are also present.

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Appendix (A) – Distribution analysis of effect size.

Figure 8. Effect sizes, plotted for bins 1 through 5 and per experiment. A) Experiment 1, B) Experiment 2 - 75%High pitch group, C) Experiment 2 - 75%Low pitch group.
* $p < .05$, ** $p < .01$, *** $p < .001$.

